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Supply Chain & Manufacturing Analytics Platform Cost Reduction, Forecasting & Vendor Performance Optimization

EY

Shape the future with confidence

Business Context

A mid-sized to large manufacturing organization is facing rising logistics costs, frequent inventory stockouts, supplier delivery delays, and production downtime, leading to reduced profitability and service levels. The management wants to build a centralized Supply Chain & Manufacturing Analytics Platform to gain end-to-end visibility across procurement, inventory, production, and logistics, improve demand forecasting accuracy, and proactively manage supplier and operational risks. Students will act as analytics consultants to design and implement an end-to-end supply chain analytics solution.

Project Objectives

1. Build a supply chain data warehouse in SQL Server
2. Design ETL pipelines for raw procurement, inventory, and logistics data
3. Write complex SQL queries for supply chain performance analysis
4. Develop Power BI dashboards for operational and executive stakeholders
5. Perform statistical analysis for operational decisionmaking
6. Build predictive models using Python for forecasting and risk identification
7. Integrate predictive model outputs into BI dashboards

8.Provide actionable business recommendations for cost reduction and efficiency improvement

Dataset Description

suppliers.csv – supplier details, location, ratings
purchase_orders.csv – purchase order dates, quantities, promised vs actual delivery inventory.csv – SKU-level stock, warehouse, reorder levels production.csv – production output, downtime hours, downtime reasons logistics.csv – shipment cost, transport mode, transit delays sales_forecast.csv – historical demand and forecasted demand
calendar.csv – date dimension (2019–2024)

Phase 1: Database Design & Data Loading

Design staging, dimension, and fact tables in SQL Server using a star schema optimized for supply chain analytics. Load CSV files using Import Wizard or BULK INSERT.

Deliverables: ER diagram, SQL DDL scripts, loaded staging tables

Phase 2: ETL & Data Transformation

Clean and standardize procurement, inventory, and logistics data by handling missing values, removing duplicates, and validating delivery and cost fields.
Derive key metrics such as inventory turnover, order fill rate, supplier on-time delivery percentage, logistics cost percentage, production downtime ratio, and lead time variability.
Load transformed data into dimension and fact tables.
Deliverables: ETL scripts and populated data warehouse

Phase 3: SQL Analytics

Write complex SQL queries using CTEs, window functions, subqueries, and ranking logic.

Business SQL questions include:

1.Identification of suppliers contributing to maximum delivery delays.

2.Calculate average delivery delay per supplier.

3.Month-over-month logistics cost and efficiency.

4.Supplier performance ranking based on delivery.

5.Identify top 3 delayed suppliers per month.

6.Calculate average shipment cost per transport

mode.

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Deliverables: SQL query file and analytical output views.

Phase 4: Power BI Dashboards

Dashboards:

1. Show supplier-wise average delivery delay.
2. Forecast future inventory levels using historical demand trends.
3. Calculate rolling 3-month Shipment cost.
4. Create drill-through pages for SKU and Supplier.
5. Create an On-Time Delivery % measure.
6. Which suppliers contribute most to inventory imbalance.
7. Rank suppliers using composite KPIs.
8. Quantify impact of logistics delays on production.
9. Analyze supplier dependency across SKUs.

Deliverables: PBIX file and dashboard screenshots.

Phase 5: Statistical Analysis

Conduct hypothesis testing to assess the effect of supplier delays on production downtime, use regression modeling to predict logistics costs based on transport mode, shipment delays, and order volume, perform correlation analysis to examine the relationship between lead time and stock-out frequency, and apply outlier detection to identify abnormal logistics cost spikes or extreme downtime incidents.

Deliverables: Jupiter notebook with statistical findings and interpretations

Phase 6: Predictive Analytics

Use Case 1: Demand Forecasting (Regression)
Use Case 2: Supplier Delay Prediction (Classification)
Use Case 3: Stock-Out Risk Prediction

Build models using Logistic Regression, Random Forest, and time series techniques. Evaluate models using RMSE, MAPE, accuracy, and business impact metrics.
Deliverables: Python notebooks and model evaluation results.

Phase 7: Integration Back to BI

Push forecast results, supplier risk scores, and stockout probabilities back into SQL Server and visualize theminPowerBIdashboardstoenableearlywarning systems and proactive decision-making

Phase 8: Business Recommendations

~~Propose data-driven actions for supplier rationalization~~ and renegotiation, inventory optimization through improved reorder and safety stock policies, logistics route and cost optimization, forecast-driven production planning, and implementation of early warning systems with quantified business impact.

Final Deliverables

SQL scripts, ETL documentation, Power BI dashboards, Python notebooks, final PPT presentation, and executive summary. **Evaluation Rubric**

Data Modelling & ETL – 20%
SQL Complexity – 15%
Dashboard & Storytelling – 20%
Predictive Modelling – 25%
Business Insights – 15%
Presentation – 5%

Learning Outcomes

Students will demonstrate end-to-end supply chain analytics pipeline design, strong SQL and BI capabilities, application of forecasting and machine learning in operations, and the ability to translate

analytical insights into actionable business decisions for stakeholders.